

Performance Testing

Date	11 July2026
TeamID	XXXXXX
Project Name	EduGenie–Google Gemini Powered Educational Assistant
Maximum Marks	5Marks

Step1:Testing Overview

Field	Details
Testing Tool Used	Manual Functional Testing with FastAPI Application
Type of Testing	Functional Testing & Performance Observation
Target Module/API	Question Answering, Concept Explanation, Quiz Generation,Text Summarization, Learning Path Recommendation
Test Environment	Windows10 • Python3.14 • FastAPI • HTML/CSS • Gemini API
Test Date	15March2026

Step2:Test Scenarios

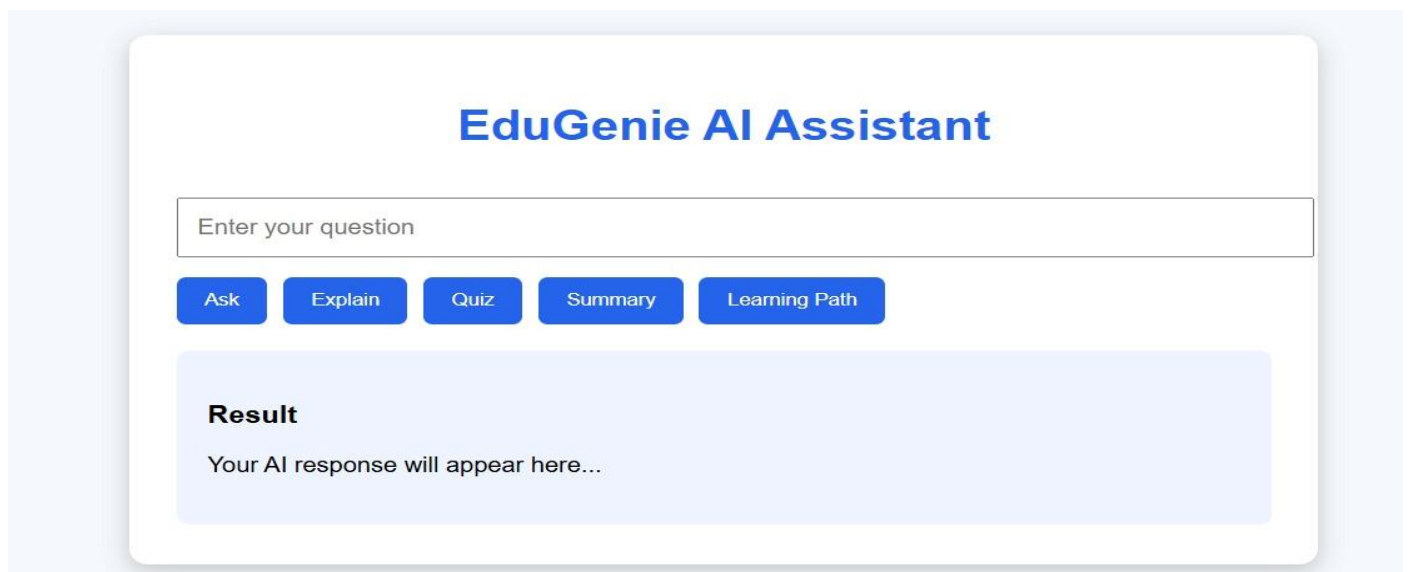
S.No	Scenario	Users	Duration	Expected Outcome
1	Launch EduGenie Application	1	3-5 sec	Homepage loads successfully
2	Ask Educational Question	1	4-8 n sec	AI returns an accurate answer with education context

3	Generate Concept Explanation	1	5-8 sec	Simplified explanation is displayed correctly
4	Generate AI Quiz	1	5-10 sec	Topic-based MCQs are generated successfully
5	Summarize Educational Text	1	4-8 sec	Concise summary is generated
6	Generate Learning Path	1	6-10 sec	Personalized learning road map is displayed
7	Navigate Between Modules	1	Lessthan2 sec	Smooth page transitions without errors

Step3:Observations

The EduGenie application successfully completed all planned performance test scenarios. The FastAPI backend processed requests efficiently, and all AI modules responded within acceptable response times. Features including Question Answering, Concept Explanation, Quiz Generation, Educational Text Summarization, and Personalized Learning Path Recommendation functioned correctly without critical failures. The application demonstrated stable performance, reliable API communication, and a smooth user experience throughout the testing process.

Step4:Screenshots/Evidence



Home Page

Welcome to EduGenie 🧠✨

Your personal AI tutor for learning support!

Ask EduGenie a Question:

what is cloud computing?

Get Answer

Answer:

Cloud computing is the delivery of computing services like servers, storage, databases, networking, software, and analytics over the internet ("the cloud") to offer faster innovation, flexible resources, and cost savings.

Question Answering

Need an Explanation?

Pandas in Python

Explain

Explanation:

Pandas is a powerful Python library used for data manipulation and analysis. It provides data structures like Series (1D) and DataFrame (2D) to store and analyze data efficiently. With Pandas, you can easily load, clean, transform, analyze, and visualize data using simple functions. It is widely used in data science, machine learning, and data analysis projects.

Explanation Output

Summarize a Paragraph:

The Industrial Revolution, which began in the late 18th century, marked

Summarize

Summary:

The Industrial Revolution changed the world by shifting it from farming to factories. New machines, like James Watt's steam engine, made things faster and easier to produce. While this boosted economies, it also created problems like bad working conditions, child labor, pollution, and overcrowded cities. Even with these problems, the Industrial Revolution shaped the modern world, affecting how we travel, communicate, work, and trade.

Educational Text Summary

Generate a Quiz:

Popular AI tools

Generate Quiz

Quiz:

Q1: Which of the following is a popular AI tool used for conversational AI and chatbots?

- ☐ TensorFlow
- ☐ Midjourney
- ☐ OpenAI ChatGPT
- ☐ Figma

Check Answer

AI-Generated Quiz

Need a Learning Path?

Get a personalized roadmap to learn any topic step-by-step.

DSA (Data Structures and Algorithms) Roadmap

Get Learning Path

Learning Path:

Here's a step-by-step DSA (Data Structures and Algorithms) Roadmap:

- 1 Learn the Basics**
Understand time complexity (Big O), variables, loops, conditionals, and functions.
- 2 Data Structures**
Learn Arrays, Strings, Linked Lists, Stacks, Queues, Hash Tables.
- 3 Algorithms**
Study Searching, Sorting, Recursion, Backtracking, Divide and Conquer, Greedy Algorithms.
- 4 Advanced Topics**
Learn Trees, Binary Search Trees, Heaps, Graphs, Tries, Dynamic Programming.
- 5 Practice Problems**
Solve problems on platforms like LeetCode, HackerRank, Codeforces, and GeeksforGeeks.
- 6 Build & Apply**
Work on real-world projects and apply DSA in problem-solving and interviews.

Consistency and practice are the keys to mastering DSA. Keep learning and building! 🚀

Personalized Learning Path Recommendation